

# Extracellular matrix viscoelasticity regulates spatiotemporal mammary branching morphogenesis

UC SANTA BARBARA

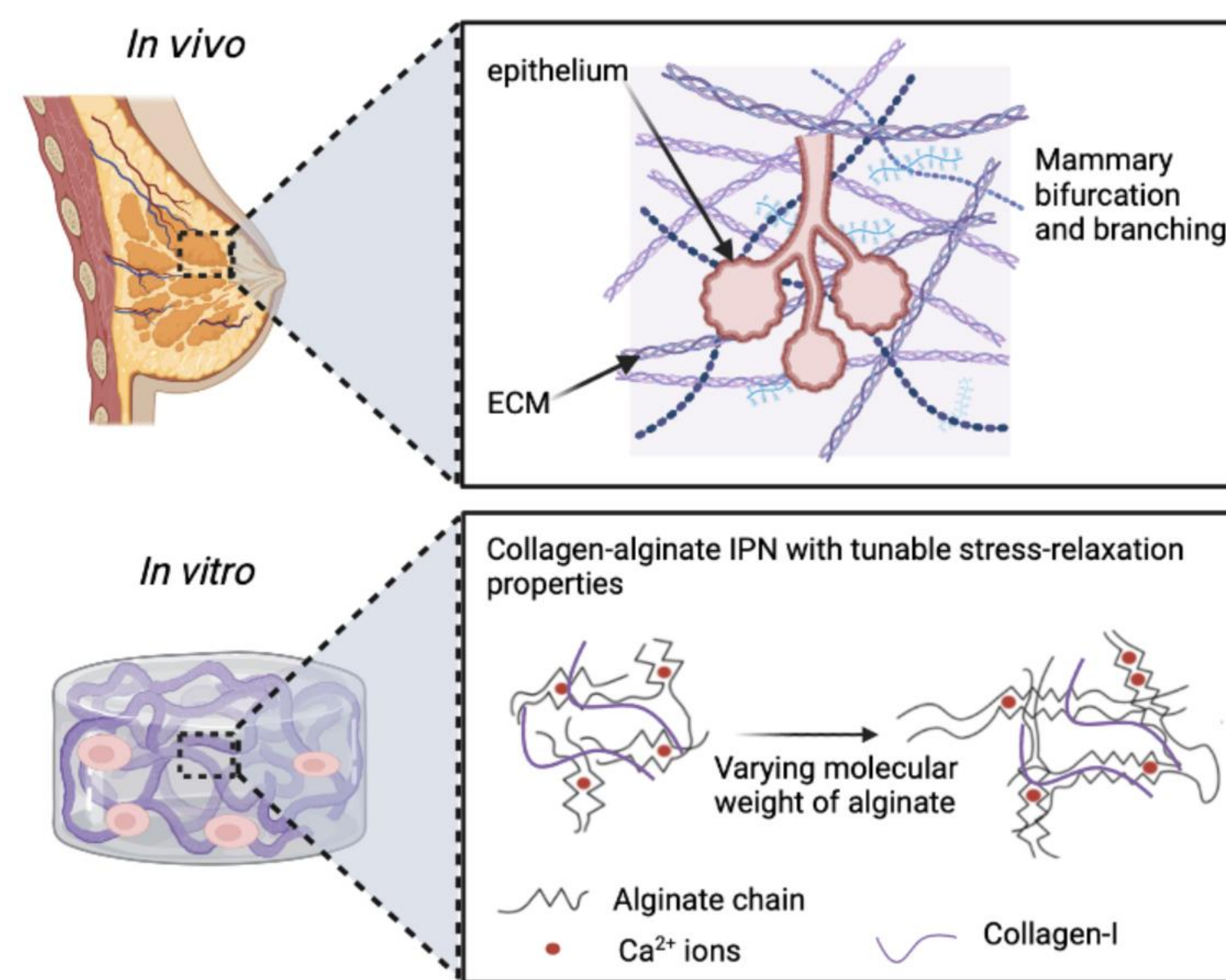
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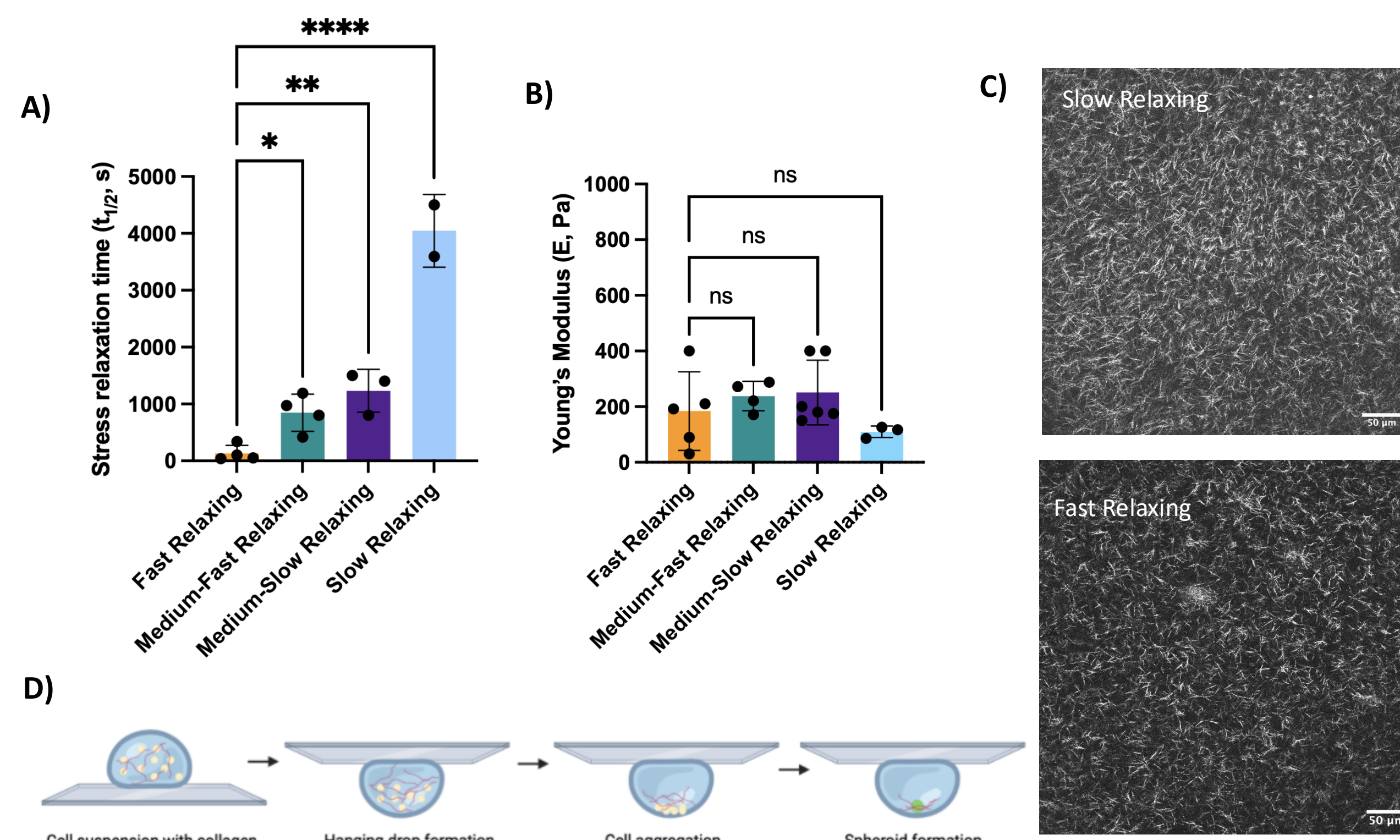
## The effect of viscoelasticity on tissue organization is unknown

Branching morphogenesis is a fundamental process that occurs during the development of mammary tissue, and several signaling pathways that regulate mammary branching are disrupted during the progression of breast cancer. Mammary gland development is influenced by mechanical and chemical cues within its *in vivo* microenvironment, and conventionally, tissue development has been studied with *in vivo* or *ex vivo/in vitro* systems where mechanical properties are poorly controlled in space and time, lack tunability, or do not mimic the native extracellular matrix (ECM) mechanics. Here, we engineered a collagen-alginate interpenetrating network (IPN) to recapitulate the mammary *in vivo* microenvironment and determine how ECM viscoelasticity regulates tissue branching in mammary development.



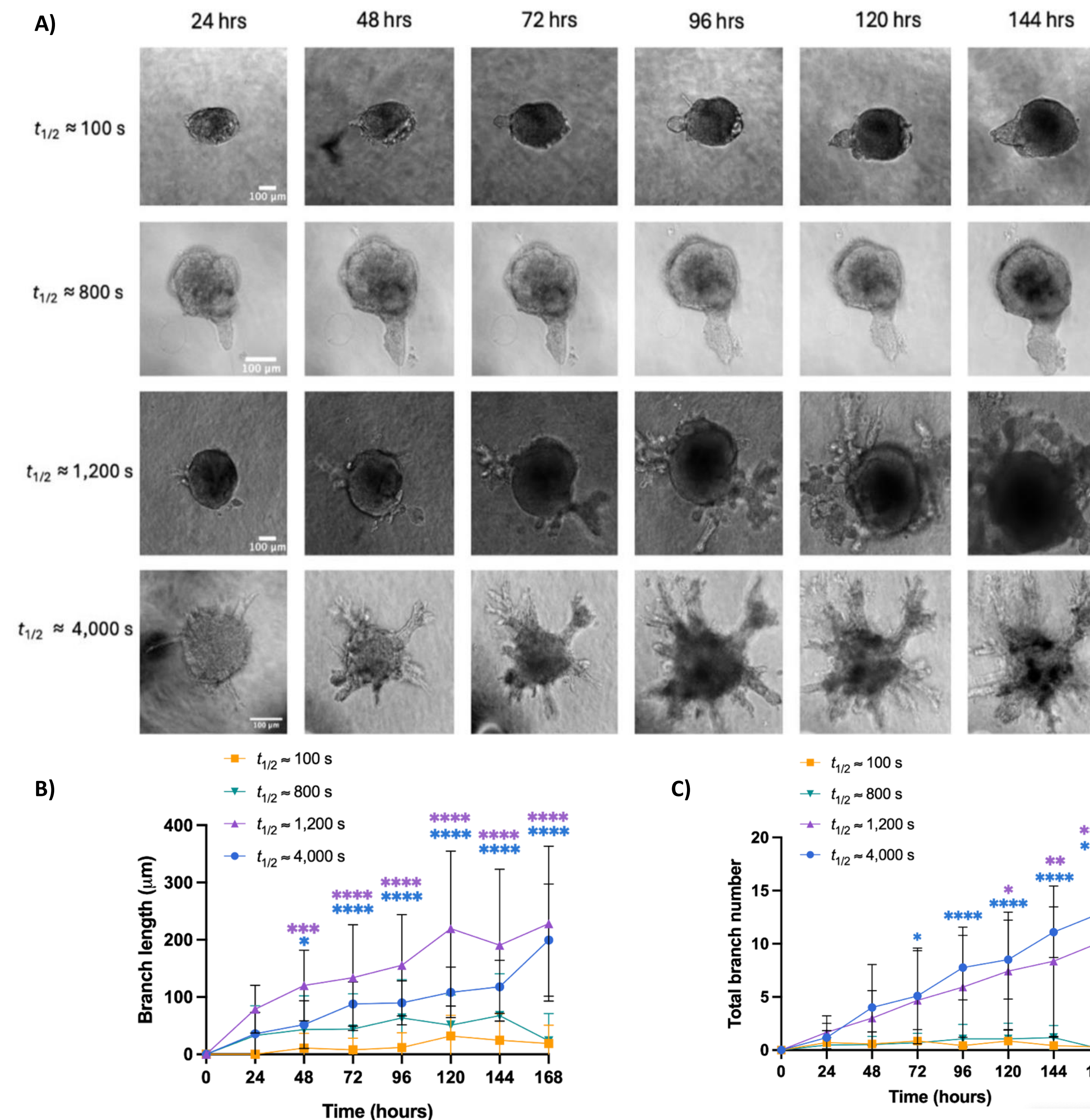
**Figure 1:** Schematic depicting how mammary gland development is influenced by mechanical and chemical cues within its *in vivo* microenvironment, and 3D collagen-alginate interpenetrating networks (IPNs) enable independent tuning of the viscoelasticity of the matrix. Figure created in BioRender.

## 3D collagen-alginate IPNs enable independent tuning of the viscoelasticity of the matrix



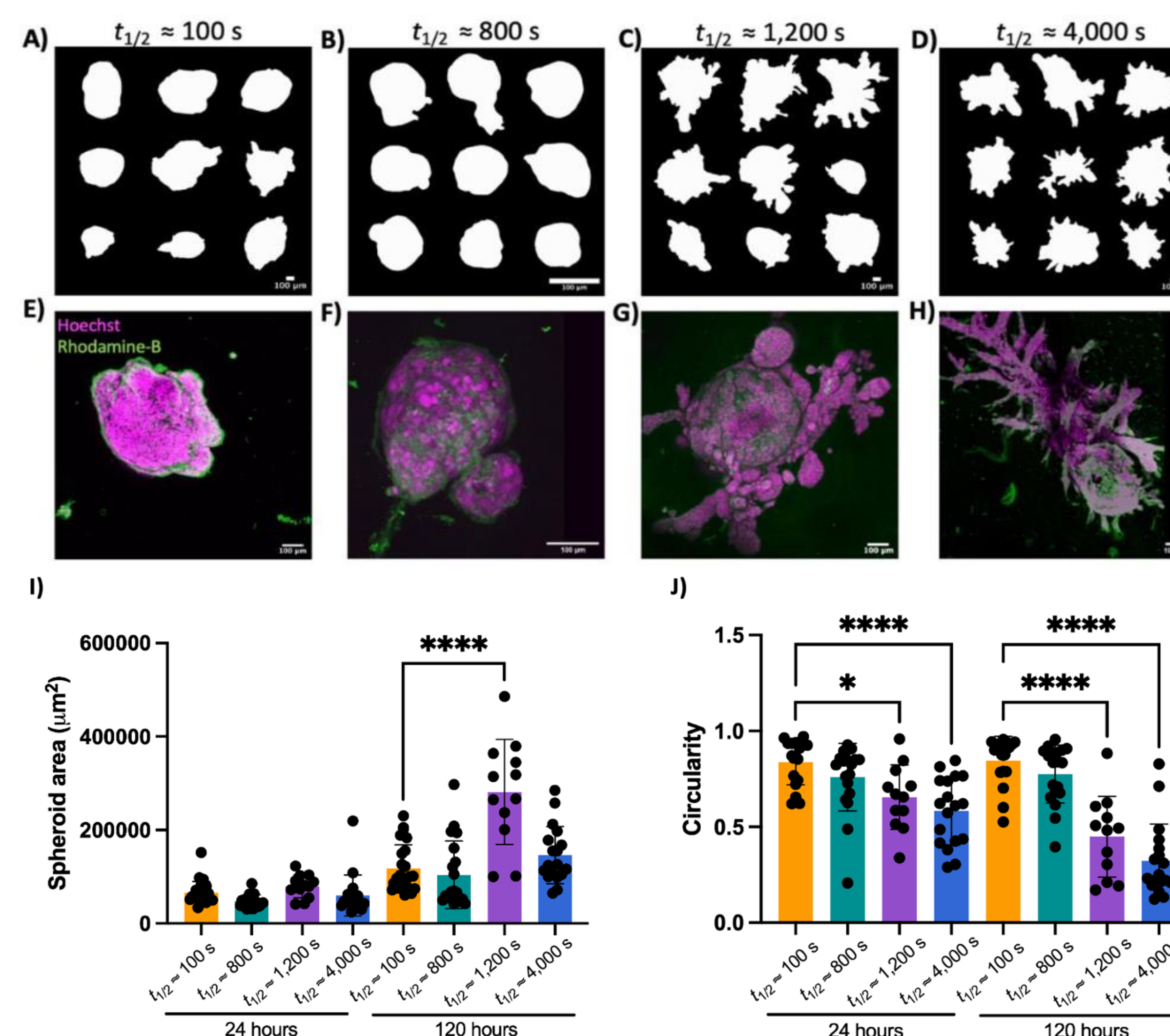
**Figure 2:** (a) Molecular weight of alginate can be tuned to fabricate matrices with varying stress-relaxation times. (b) Rheology of hydrogels reveals that hydrogels with varying stress-relaxation times have matrix stiffnesses that are not significantly different. (c) Confocal reflectance microscopy reveals that slow relaxing matrices and fast relaxing matrices have similar collagen microarchitectures. (d) MCF10A spheroids were prepared via the hanging-drop method and encapsulated in matrices after 48 hours. Image created in BioRender.

## Mammary epithelial spheroid morphology varies with stress-relaxation rate



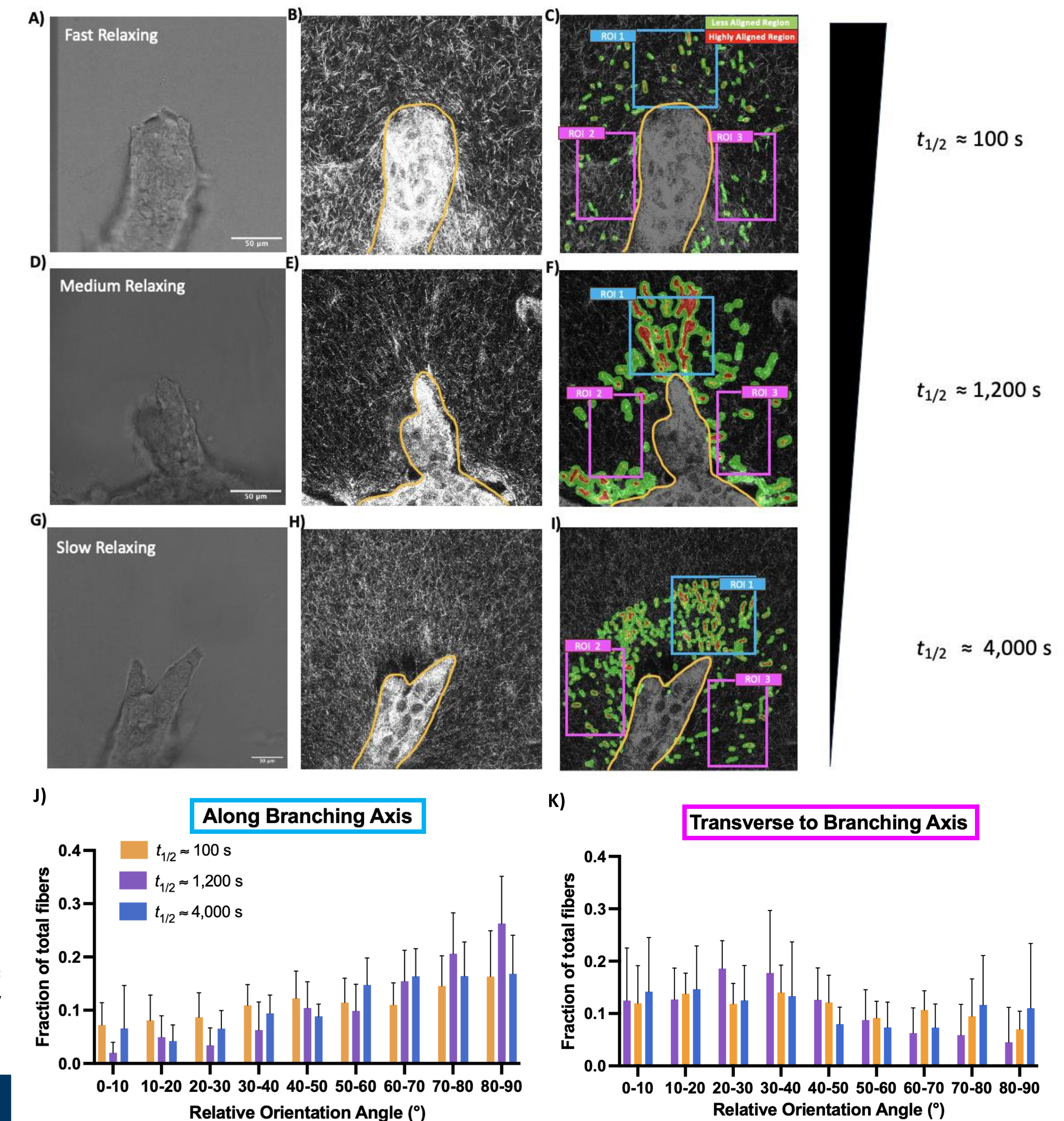
**Figure 3:** (a) Brightfield images of MCF10A spheroids in 3D slow stress-relaxing matrices ( $t = 1200$  s,  $t = 4000$  s) and fast stress-relaxing matrices ( $t = 100$  s,  $t = 800$  s) over 7 days. (b) Quantification of primary branch length demonstrates a faster branch growth rate in slow stress-relaxing matrices. (c) Quantification of the number of branches demonstrates more branches are formed in slow stress-relaxing matrices.

## Slow stress-relaxing matrices enhance spheroid area



**Figure 4:** (a-d) Representative outlines of MCF10A spheroid morphologies in varying stress-relaxing matrices after 120 hours in culture. (e-h) Confocal images of a MCF10A spheroid in slow and fast stress-relaxing matrices (magenta=nuclei, green=cell membrane). (i) MCF10A spheroid area in slow and fast stress-relaxing matrices after 120 hours (n=18 and 21, respectively). (j) MCF10A circularity in slow and fast stress-relaxing matrices after 120 hours (n=18 and 21, respectively).

## Slow stress-relaxing matrices promote collagen fiber alignment along branching axis



**Figure 5:** (a) Brightfield image of a spheroid branch in a fast stress-relaxing matrix ( $t = 100$  s) (b) Confocal reflectance microscopy reveals collagen fiber network surrounding a MCF10A spheroid branch in a fast stress-relaxing matrix ( $t = 100$ s), where the yellow outline indicates the spheroid boundary. (c) Heatmap demonstrating regions of high and low collagen fiber alignment in a fast stress-relaxing matrix (red=high alignment, green=low alignment). ROI1 represents quantifications along the branching axis and ROI 2 and 3 represent quantifications transverse to the branching axis. (d-f) Representative images of collagen fiber alignment for a branch in a medium-slow stress-relaxing matrix ( $t = 1200$  s). (g-i) Representative images of collagen fiber alignment for a branch in a slow stress-relaxing matrix ( $t = 4000$  s). (j) Quantification of relative orientation angle demonstrates that collagen fibers are highly aligned in slow-relaxing matrices ( $t=1200$  s) along the branching axis and (k) randomly aligned along the transverse axis. Collagen alignment was quantified via CurveAlign. 9 spheroids were imaged per gel condition.  $p < .0001$  for all groups.

## Conclusions and Acknowledgments

- 3D alginate-collagen interpenetrating networks (IPNs) mimic *in vivo* microenvironment and enable independently tuning of the stress-relaxation of the matrix.
- Extracellular matrix viscoelasticity regulates spatiotemporal epithelial tissue organization within human mammary epithelial cells

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